

# RECYM: A Multi-Feeder CYMDIST Workflow for Demand Allocation, Spot-Load Studies, and Situational–Projected Load-Flow Comparison

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**Abstract**—Distribution utilities must repeatedly correct network models, allocate measured headroom demand, connect prospective concentrated loads, and compare base versus future load-flow cases inside commercial tools such as CYMDIST. Manual campaigns are error-prone, hard to audit, and difficult to reproduce across dozens of feeders. This paper presents RECYM, a multi-feeder software suite that couples CYMDIST/CymPy and COM engines with a seven-step single-page application and a batch pipeline. The workflow enforces a model-quality gate, maps large-customer energy and contracted power onto secondary distribution transformers, allocates residual demand by energy (kWh) against a measured headroom total, inserts locked three-phase spot loads on existing nodes, and runs dual load-flow scenarios that disconnect or reconnect the prospective load. On Electro Dunas feeder PA217 (22.9 kV; 1324 nodes; 253 loads), RECYM preserved headroom balance at 9537.88 kW, updated ten secondary transformers with energy-to-kWh and power-to-kW locked injections, and connected a 1420 kW spot load for situational–projected comparison, with zero diagnostic problems on the quality board after corrections. The architecture, coupling rules, and reproducible artifacts are documented so peer utilities can replicate the campaign on additional feeders.

**Index Terms**—CYMDIST, demand allocation, distribution networks, load flow, power system modeling, software tools, spot load.

## I. INTRODUCTION

Planning and connection studies on medium-voltage distribution feeders require a trustworthy digital twin: correct connectivity and equipment data, demand that matches metered headroom, explicit treatment of large customers, and a transparent comparison between the present operating point and a projected case that includes a new concentrated load [1], [2]. Commercial packages such as CYMDIST provide load allocation and load-flow engines, but production campaigns still rely on long sequences of manual clicks, spreadsheets, and ad-hoc scripts.

Open platforms emphasize flexible simulation APIs [3], while vendor documentation covers allocation and load-flow algorithms in isolation [4], [5]. What remains scarce is an end-to-end operational workflow—diagnostics, large-customer injection, kWh-based residual allocation, locked spot-load insertion, dual-scenario load flow, and delivery-report generation—bound to a multi-feeder configuration and an auditable user interface.

This paper presents RECYM. The contributions are: (i) a layered SPA/API/pipeline/CymPy–COM architecture for one

feeder or batch modes; (ii) a seven-step campaign with coupling rules that prevent incorrect mixing of allocation, spot load, and situational/projected cases; and (iii) an industrial case study on feeder PA217 demonstrating headroom-consistent allocation, SED updates, a 1420 kW spot load, and reproducible artifacts.

## II. RELATED WORK

Distribution modeling fundamentals are well established [1], [2]. Load allocation to match metered feeder totals is standard utility practice [4]. Compensation-based power flow for weakly meshed networks underpins many engines [6]. OpenDSS [3] and pandapower [7] automate research studies; RECYM instead orchestrates CYMDIST engines for industrial campaigns.

## III. SYSTEM ARCHITECTURE

### A. Layered Design

RECYM comprises: (1) a React SPA with seven operator panels; (2) a FastAPI job layer with SSE progress and a Flask bridge; (3) domain pipelines (diagnostics, customers, allocation, spot load, load flow, reports, optional optimization); (4) CymPy and COM adapters; and (5) external .xlsx/.mdb/Excel data. Fig. ?? (placeholder) shows the stack.

### B. Multi-Feeder Configuration

Global paths live in `settings.json`; each feeder JSON stores `network_id`, study path, workbooks, and `data/output/feeders/<ID>/. Writes require a resolved study path and dry_run=false.`

## IV. PROPOSED WORKFLOW

### A. Seven-Step Campaign

Table I summarizes the campaign. After connecting a spot load, residual allocation must not be re-run; only load flow follows. Situational versus projected differs solely by disconnecting or reconnecting the prospective spot load.

### B. Spot-Load Phase Split

For three-phase demand  $P$  and  $Q$  at an existing node,

$$P_\phi = \frac{P}{3}, \quad Q_\phi = \frac{Q}{3}, \quad \phi \in \{A, B, C\}. \quad (1)$$

TABLE I  
RECYM OPERATOR CAMPAIGN

| Step | Panel      | Function                               |
|------|------------|----------------------------------------|
| 1    | Context    | Bind DB/study; save headroom $P, Q$    |
| 2    | Quality    | NetworkDiagnostic; corrections; gate   |
| 3    | Customers  | SED EA→kWh, Pot→kW; allocation         |
| 4    | Spot load  | Existing node; $P/3$ per phase; Locked |
| 5    | Load flows | Situational (off) / projected (on)     |
| 6    | Reports    | Meta OCR + Word/PDF package            |
| 7    | Suite      | Optional equipment optimization        |

## V. CASE STUDY SETUP

Electro Dunas feeder PA217 (NET\_2030\_179\_PA217),  $V_{ll} = 22.9\text{ kV}$ , has 1324 nodes, 1324 sections, and 253 loads. Headroom:  $P = 9537.88\text{ kW}$ ,  $Q = 2587.65\text{ kvar}$ ,  $S = 9882.67\text{ kVA}$  (loading factor 65.05%). Prospect load: 1420 kW at node NODE\_1080\_320357\_MI-17145\_2. Software stack: CYMDIST 9.2, CymPy, COM, RECYM SPA v6.

## VI. RESULTS AND DISCUSSION

After corrections, the quality board reported zero problems. Ten SED loads were written Connected with EA→Consumo and Pot→kW Locked; one customer was excluded (Disconnected); one row lacked SED mapping. Allocation validation:  $\sum P = 9537.87\text{ kW}$  versus headroom 9537.88 kW ( $d = 0.0\%$ ). The 1420 kW spot load uses equal phase split ( $P_\phi = 473.33\text{ kW/phase}$ ). Integral validation for PA217 returned PASS (21 September 2026). Load-flow  $V_{\min}$ /loss scalars must be frozen from a non-dry-run export before camera-ready tables. Limitations include Windows COM dependence, single-session assumptions, and utility-specific workbook mapping.

## VII. CONCLUSION

RECYM orchestrates CYMDIST for multi-feeder demand and connection studies via a quality gate, SED injection, kWh allocation, locked spot loads, and situational–projected load flows. Future work includes multi-feeder LF KPI packs and production use of optional equipment optimization.

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